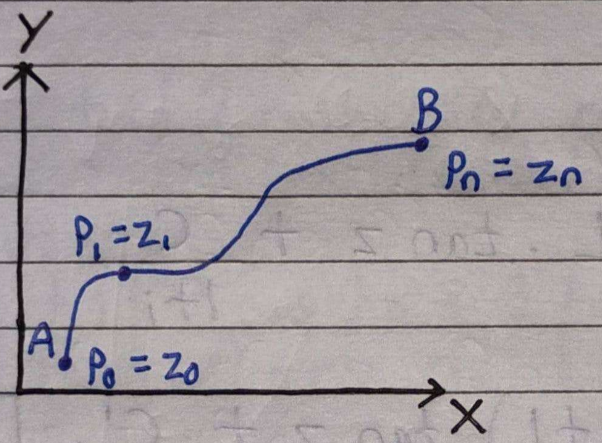
Line Integrals in Complex Plane

Let, $F(z)$ be a continuous function of the complex variable $z = x + iy$, defined at all points along a curve C with end-points A & B. Divide the curve C into n sub-arcs at the points:

$$A = P_0(z_0),\ P_1(z_1),\ P_2(z_2), \ldots, P_n(z_n) = B$$



Let, z_{i-1} & z_i be the endpoints of the i-th sub-arc and define the change in z as:

$$\delta z_i = z_i - z_{i-1}$$

Let, ξ_i be any point chosen on the sub-arc $P_{i-1}P_i$, then the limit of the sum as $n \to \infty$ and $\delta z_i \to 0$ is defined as:

$$\sum_{i=1}^{n} F(\xi_i)\,\delta z_i \quad , \text{when } n \to \infty \ \& \ \delta z_i \to 0.$$

If this limit exists, it is called the line integral of $F(z)$ along the curve C i.e. $\int_C F(z)\,dz$.

In case points P_0 & P_n coincide (i.e. $A=B$), then the curve C forms a closed curve. The integral along such a closed path is called a Contour Integral, which is denoted by:

$$\oint_C F(z)\, dz.$$

To evaluate the integral using real variables, substitute $F(z) = u + iv$ and $z = x + iy \Rightarrow dz = dx + i\,dy$.

$$\int_C F(z)\, dz = \int_C (u+iv)(dx + i\,dy)$$

$$= \int_C (u\,dx + iu\,dy + iv\,dx - v\,dy)$$

$$= \int_C (u\,dx - v\,dy) + i\int_C (v\,dx + u\,dy).$$

9. Find the value of the integral $\int_C \{(x+y)\,dx + x^2y\,dy\}$

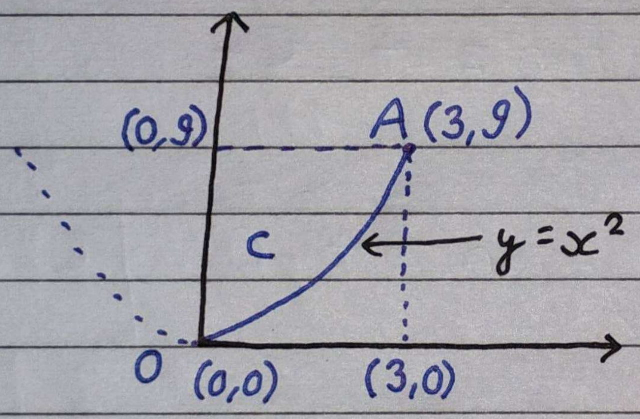
i. along $y=x^2$ having $(0,0)$ and $(3,9)$ as end points.

ii. along $y=3x$ between the same points.

Sol^n:

Given, $\int_C \{(x+y)\,dx + (x^2y)\,dy\}$

i. along $y=x^2$ joining $(0,0)$ & $(3,9)$,



$y = x^2$

$\Rightarrow dy = 2x\,dx$

$$\int_C \{(x+y)\,dx + (x^2y)\,dy\}$$

$$= \int_0^3 \{(x+x^2)\,dx + (x^2 . x^2)\,dx\,(2x)\}$$

$$= \int_0^3 (x+x^2+2x^5)\,dx$$

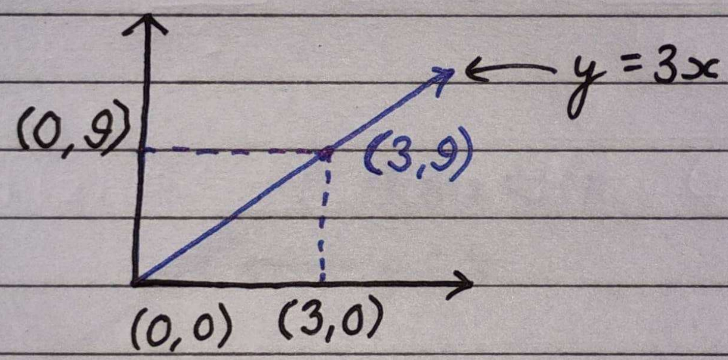
$$= \left[\frac{x^2}{2} + \frac{x^3}{3} + \frac{\cancel{2}x^6}{\cancel{6}\,3} \right]_0^3$$

$$= \frac{9}{2} + \frac{\cancel{27}^{9}}{\cancel{3}} + \frac{\cancel{729}^{243}}{\cancel{3}}$$

$$= \frac{9}{2} + 9 + 243$$

$$= \frac{513}{2}.$$

ii. along $y = 3x$ between $(0,0)$ & $(3,9)$,



$$y = 3x$$

$$\Rightarrow dy = 3\,dx$$

$$\int_C \{(x+y)\,dx + (x^2 y)\,dy\}$$

$$= \int_0^3 \{(x + 3x)\,dx + (x^2 . 3x)(3\,dx)\}$$

$$= \int_0^3 (4x + 9x^3)\,dx$$

$$= \left[\frac{4x^2}{2} + \frac{9x^4}{4} \right]_0^3$$

$$= \left[2x^2 + \frac{9x^4}{4} \right]_0^3$$

$$= 18 + \frac{729}{4}$$

$$= \frac{801}{4}.$$